EXP NO: 19 EXPERIMENT TO UNDERSTAND ARRAY FUNCTION IN DATA SCIENCE

AIM:

Understand array function using Numpy library.

PYTHON CODE:

```
import numpy as np
array = np.random.randint(1, 100, 9)
array = np.array([83, 25, 19, 47, 62, 15, 96, 39, 51])
print(array)
print(np.sqrt(array))
print(array.ndim)
new_array = array.reshape(3, 3)
print(new_array)
print(new_array.ndim)
print(new_array.ravel())
newm = new_array.reshape(3, 3)
print(newm)
print(newm[2, 1:3])
print(newm[1:2, 1:3])
print(new_array[0:3, 0:0])
print(new_array[0:2, 0:1])
print(new_array[0:3, 0:1])
print(new_array[1:3])
```

OUTPUT:

```
[83 25 19 47 62 15 96 39 51]
[9.11043358 5.         4.35889894 6.8556546  7.87400787 3.87298335
 9.79795897 6.244998   7.14142843]
1
[[83 25 19]
 [47 62 15]
 [96 39 51]]
2
[83 25 19 47 62 15 96 39 51]
[[83 25 19]
 [47 62 15]
 [96 39 51]]
[39 51]
[[62 15]]
[]
[[83]
 [47]]
[[83]
 [47]
 [96]]
[[47 62 15]
 [96 39 51]]
```

RESULT:

Thus the code is executed successfully.